

Design of Rock Bolting Systems for Underground Excavations

INTRODUCTION

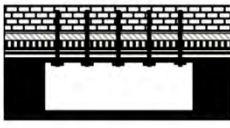
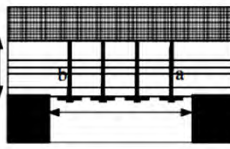
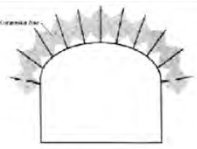
- Rock bolting has been widely used in coal and hard rock mining, underground hydro-electric schemes, tunnelling, and other rock structures in mining, civil and military engineering.
- With the development of various analytical theories, new techniques such as finite element, boundary element, distinct element, probabilistical and statistical methods have been developed and used in rock bolting design.
- The design methods used in practice change from place to place and from time to time due to variation in geological conditions and underground structures.

INTRODUCTION

- A successful operation of rock bolting system depends on:
 - basic data, from both the laboratory and the field;
 - correct selection of failure criteria and rock bolting material; and
 - appropriate selection of design approach.

INTRODUCTION

- The approaches used in the design of rock bolting can be broadly divided into three categories:
 - empirical approach,
 - analytical approach, and
 - a combined approach.

Theory	Description	Shape	Comments	Reference
Suspension	The dead weight load of the strata transferred between the bolt head and the anchor.		The upper layer should be strong enough to anchor the bolts	Peng (1984) Stillborg (1986)
Beam building	Bolts bind the strata layers together, which prevents or greatly reduce the movements. The frictional effect generated by bolt pretensioning increases the shear strength between the layers.		The tensile failure is prevented because of the increase in binding strength and stiffness effect. However, failure may still occur by shearing at the two ends. Hence bolts a and b are most effective in this method.	Peng (1984) Panek (1956) Xiu (1997) Snyder (1983) Stillborg (1986)
Keying effect	When the roof strata are highly fractured and blocky, or planes of weakness intersect the immediate roof strata, keying effect roof support may be used. Roof bolting provides significant frictional forces along fractures, cracks, and weak planes		Confinement is provided by way of tensioned roof bolts, and then the strata material will be locked or keyed together.	Peng (1986)
Arching Action	The main aim of the arching theory is to increase the value of compressive stresses in the roof so that the tensile stress is ignored and the shear resistance increased. Arching provides very strong roof profile, which is one of the strongest roof profiles		To support the weakened zones between bolts, the use of wire mesh and shotcrete is recommended	Wagner (1997)

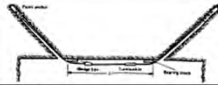
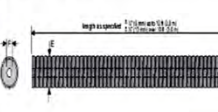
Types of roof bolts (Peng, 1984)

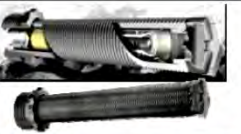
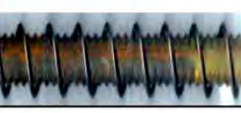
Types of bolt	Types of anchor	Suitable strata type	Comments
Point-anchored bolts (tensioned)	Slot-and-wedge	Hard rock	Used in the early stages
	Expansion shell		Most commonly used in the U.S.A.
	Standard anchor	Medium-strength rock	
	Bail anchor	Soft rock	
	Explosive set	Lower-strength rock	Limited use
	Resin grout		Increasing usage recently
	Pure point anchor	All strata especially for weak rock	Resin length ≤ 24 in.
	Combination system		Resin length ≥ 24 in.
Full-length-grouted bolt (untensioned)	Combination anchor (expansion shell and no mix resin)	Most strata	Good anchorage with "no mix resin"
	Cement	Most strata	Disadvantages: 1. Shrinkage of cement 2. Longer setting time
	Perfo		
	Injection		
	Cartridge		
	Resin	All strata	Increased use recently especially for weak strata
	Cartridge		

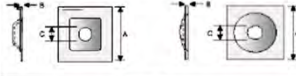
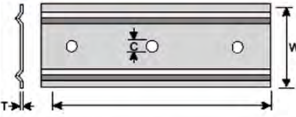
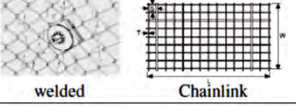
Types of roof bolts (Peng, 1984)

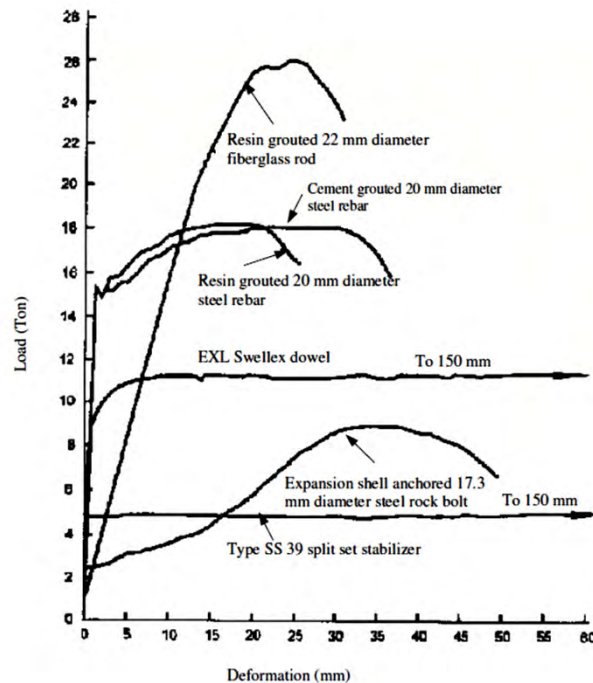
Roof truss	Expansion shell	Adverse roof	Recommended for use at intersections and/or heavy pressure area
Cable sling	Cement anchor and full-length fraction	Weak strata	Substitute for timber, steel or truss support
Yieldable bolt	Expansion shell	Medium-strength rock	It is an expansion-shell bolt with yielding device
Pumpable bolt	Resin	Weak strata	Complex installation
Helical bolt	Expansion shell	Most strata	
Split set	Full-length fraction	Weak strata	Cheap but need special installation equipment
Swellex bolt	Full-length holding	Water-bearing strata	Using high-pressure water to swell the steel tube

Types of bolts	Types of anchor	Applicability	Capacity (t)	Advantages	Disadvantages	Typical diagram
Point anchored bolts (tensioned)	Slot and wedge	Hard rock	10-16	• Inexpensive • Immediate support possible • High capacity in hard rocks.	• Limited to use in soft rock condition • Long-term stability is affected by slippage.	
	Expansion shell	Universal				
	Resin grout	Universal				
	Combination anchor[NIA1]	Most strata				
Fully length grouted bolt (untensioned)	Cement	Most strata	15-25	-High corrosion resistance - Durable - Consistent. -Increased use recently specially for weak rocks	-Shrinkage of cement, longer setting time -Installation is time critical -Costlier than mechanical bolt.	
	Injection					
	Cartridge Resin					
	Injection	All strata				
	Cartridge					
Seedsman (2005)						

Types of bolts	Types of anchor	Applicability	Capacity (t)	Advantages	Disadvantages	Typical diagram
Roof truss	Expansion shell	Adverse roof		Suitable at intersection and high pressure area	This Type of support should be used if bolt are not pretensioned	
Grouted Cable	Flexible	Universal	Up to 50 ton	Same as grouted bolt -Capability up to 15 m	Special procedure has to apply for tensioning	
	Bulb					
	Heiten					
Friction	Split set Swellex	Weak to moderate strata	10-14	Cheap, Simple installation - Capability of large displacement -Reusable.	- Relatively expensive - Hole diameter is critical for installation -Not very corrosion resistive.	
Threaded Fibre glass	Resin Anchor	Weak to moderate strata	Up to 33 ton	Does not prevent mucking up	Only drill bit rotary	

Types of bolts	Type of anchor	Applicability	Capacity (t)	Advantages	Disadvantages	Typical diagram
Helical rock bolt (Giralo 2005)	Fully grouted / point anchored	Weak rock	0.92 tone/in	HRB can be implemented with any size, grade and length of bar	Needs modification to the bolter hydraulic system	
Hilti onestep bolt www.hilti.com	Resin anchor	Weak to moderate rocks	>20 t	Faster bolt installation-increased working safety	Limitation of bolt length-Higher price compar	
Cone bolt (Big Bell 1999)	Resin anchor	Rock burst zone	Up to 20 t	Easy install-Effective even in high deformation	The performance heavily depends on grout specifications	
Posimix 4 (Mikula 2004)	Resin anchor	Moderate strengths	Up to 12.5 t	Constant annulus	-Spin torque for mixing resin is high -Spring can catch on mesh during insertion	

Accessories	Usage	Diagram	Comments
Face plate	To uniformly distribute the load at bolt collar.		Used with all types of bolt. Dome shape is the most common.
Anti friction washer	To reduce the friction between the nut and faceplate.		Increases the torque-tension ratio. Common for all bolt types.
Nuts and spherical seats	To prevent the bolt from pulling and applying pretension load		Common for all bolt types
W-strap	Pulled into rock surface by the bolt to conform major irregularities.		Provide a large surface confinement to any loose rock between bolts.
Wire mesh	To prevent injury to personnel and damage to equipment from small pieces of rock or spalled flakes.		Used right up to the face to avoid any accident at intervals between 1 to 1.5 m.

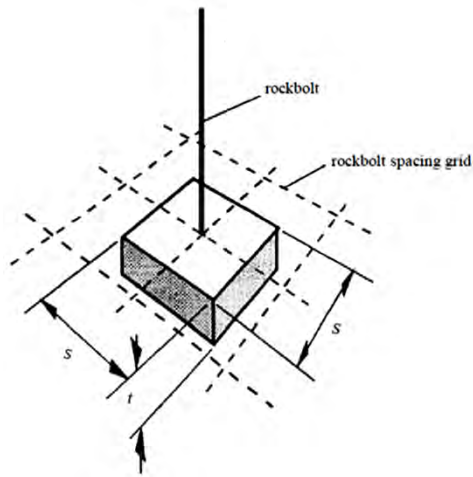


Design Methods – Empirical Approach

- The empirical assessments are based on the rock mass classification, case histories and the experience of the engineers.
- The following three empirical assessment systems are widely and successfully used for rockbolt design.
 - Empirical assessments based on the rock mass classification by Bieniawski.
 - Empirical assessments based on the Q-system by Barton et al.

Approaches to the design of support at various stages of mine development

Development stage	Information available	Design approach
Exploration and preliminary design	Location and nature of ore deposit and rudimentary information on host rock from exploration drilling and surface mapping	Use rock mass classification to obtain first estimates of support requirements
Mine design with detailed layout of permanent access excavations and stopes	Estimates of rock mass structure and strength and in situ stress conditions from mapping and measurements carried out in exploration adits and shafts	Use modified rock mass classifications and numerical models to design rockbolt patterns for permanent excavations and cable bolt patterns for stopes
Mining shafts, haulages, ramps and other permanent access ways and early stopes of mine	Detailed information on rock mass structure and strength, blast damage, and on the performance of selected support systems	Refinement of designs using elaborate numerical models and giving attention to quality control of support installation
Later years of mining and extraction of pillars	Experience from many years of refinement of excavation and support installation and, possibly, the results of displacement and load monitoring	Fine tuning of support to meet specific requirements with the use of the most advanced numerical models where required.



A roof slab of thickness t being supported by one rockbolt in a pattern spaced on a grid spacing of $S \times S$

Assuming that the unit weight of the broken rock is $\gamma = 2.7$ tonnes/m
thickness of the slab $t = 1$ m
and grid spacing $S = 1.5$ m,

The demand D on the rockbolt is equal to the weight W of the block and, if the strength or capacity of the bolt is $C = 8$ tonnes, the factor of safety
 $F = 8/6.1 = 1.3$.

- The value of the factor of safety, which is considered acceptable for a design, is usually established from previous experience of successful designs.
- A factor of safety of 1.3 would generally be considered adequate for a temporary mine opening while a value of 1.5 to 2.0 may be required for a 'permanent' excavation such as an underground crusher station.

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Design Methods – Empirical Approach

Rock Quality Designation (RQD)

- In 1964 Deere recognised a particular rock parameter based on core recovery by diamond drilling, rock quality designation (RQD), and correlated it with rock load and support requirements taking into account the influence of construction type on support design.
- This rock quality designation has come to be widely used and has been shown to be particularly useful in classifying rock masses for the selection of support systems.

Rock Quality Designation (RQD)

1 RQD (Rock Quality Designation)			RQD
A	Very poor	(> 27 joints per m ³)	0-25
B	Poor	(20-27 joints per m ³)	25-50
C	Fair	(13-19 joints per m ³)	50-75
D	Good	(8-12 joints per m ³)	75-90
E	Excellent	(0-7 joints per m ³)	90-100

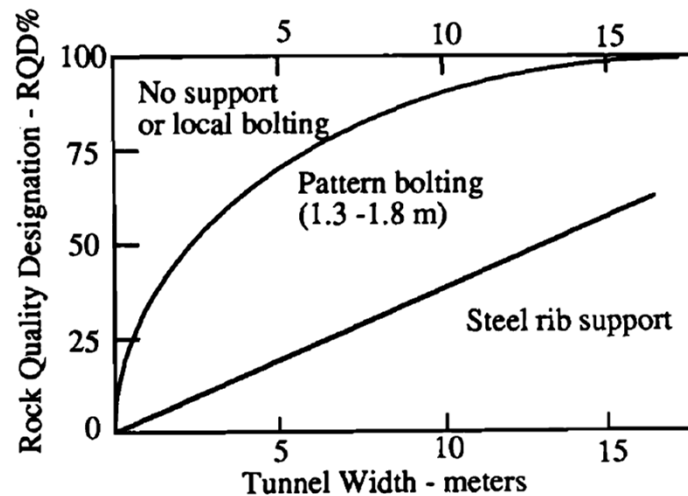
Note: i) Where RQD is reported or measured as ≤ 10 (including 0) the value 10 is used to evaluate the Q-value
 ii) RQD-intervals of 5, i.e. 100, 95, 90, etc., are sufficiently accurate

TABLE 3.4 Comparison of RQD and Support Requirements for a 6-m (20-ft)-wide Tunnel*

	No Support or Local Bolts	Pattern Bolts	Steel Ribs
Deere et al. (1970)	RQD 75–100	RQD 50–75 (1.5–1.8-m spacing) RQD 25–50 (0.9–1.5-m spacing)	RQD 50–75 (light ribs on 1.5–1.8-m spacing as alternative to bolts) RQD 25–50 (light to medium ribs on 0.9–1.5-m spacing, as alternative to bolts) RQD 0–25 (medium to heavy circular ribs on 0.6–0.9-m spacing)
Cecil (1970)	RQD 82–100	RQD 52–82 (alternatively, 40–60-mm shotcrete)	RQD 0–52 (ribs or reinforced shotcrete)
Merritt (1972)	RQD 72–100	RQD 23–72 (1.2–1.8-m spacing)	RQD 0–23

*Data interpolated from Merritt (1972) by Deere and Deere (1988).

Proposed use of **RQD** for choice of rock support system
(after Merritt, 1972)



Design Methods – Empirical Approach

Geomechanics Classification

1. Uniaxial compressive strength of rock material.
2. Rock Quality Designation (*RQD*).
3. Spacing of discontinuities.
4. Condition of discontinuities.
5. Groundwater conditions.
6. Orientation of discontinuities.

Design Methods – Empirical Approach

Geomechanics Classification

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Rock mass class	Excavation	Rock bolts (20 mm diameter, –fully grouted)	Shotcrete	Steel sets
I - Very good - rock <i>RMR</i> : 81-100	Full face, 3 m advance.	Generally no support required except spot bolting.		
II - Good rock <i>RMR</i> : 61-80	Full face, 1-1.5 m advance. Complete support – 20 m from face.	Locally, bolts in crown 3 m long, spaced 2.5 m with occasional wire mesh.	50 mm in crown where required.	None.
III - Fair rock <i>RMR</i> : 41-60	Top heading and bench 1.5-3 m advance in top heading. Commence support after each blast. Complete support 10 m from face.	Systematic bolts 4 m long, spaced 1.5 - 2 m in crown and walls with wire mesh in crown.	50-100 mm in crown and 30 mm in sides.	None.
IV - Poor rock <i>RMR</i> : 21-40	Top heading and bench 1.0-1.5 m advance in top heading. – Install support – concurrently with excavation, 10 m from face.	Systematic bolts 4-5 m long, spaced 1-1.5 m in crown and walls with wire mesh.	100-150 mm in crown and 100 mm in sides.	Light to medium ribs spaced 1.5 m where required.
V - Very poor rock <i>RMR</i> : < 20	Multiple drifts 0.5-1.5 m advance in top – heading. Install support concurrently with – excavation. –Shotcrete as soon as possible after blasting.	Systematic bolts 5-6 m long, spaced 1-1.5 m in crown and walls with wire mesh. Bolt invert.	150-200 mm in crown, 150 mm in sides, and 50 mm on face.	Medium to heavy ribs spaced 0.75 m with steel lagging and forepoling if required. –Close invert.

The most relevant outputs from a rock bolting design stand point are

- internal angle of friction and cohesion,
- unsupported stand-up time,
- maximum unsupported span and
- rock load height.

Class No.	I	II	III	IV	V
Average stand-up time	10 years for 15m span	6 months for 8m span	1 week for 5m span	10 hours for 2.5m span	30 minutes for 1m span
Cohesion of the rock mass	>400 KPa	300 - 400 KPa	200 - 300 KPa	100 - 200 KPa	<100 KPa
Friction angle of the rock mass	>45°	35° - 45°	25° - 35°	15° - 25°	<15°

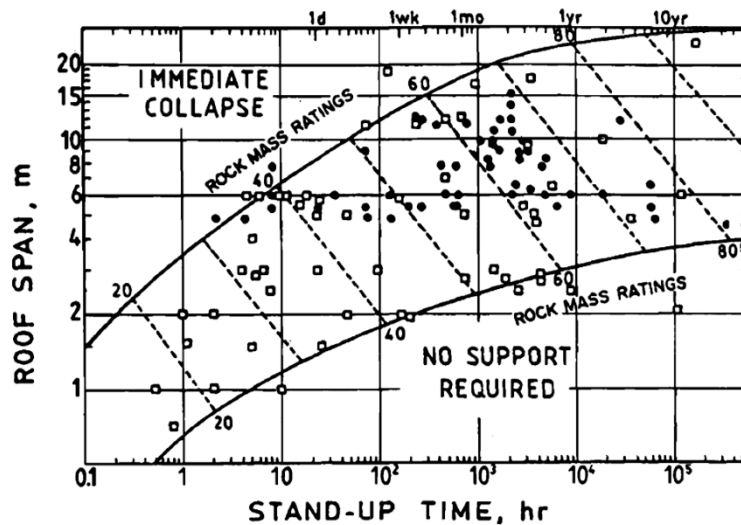


Figure 4.1 Relationship between the stand-up time and span for various rock mass classes, according to the Geomechanics Classification: output for tunneling and mining. The plotted data points represent roof falls studied: filled squares for mines, open squares for tunnels. The contour lines are limits of applicability.

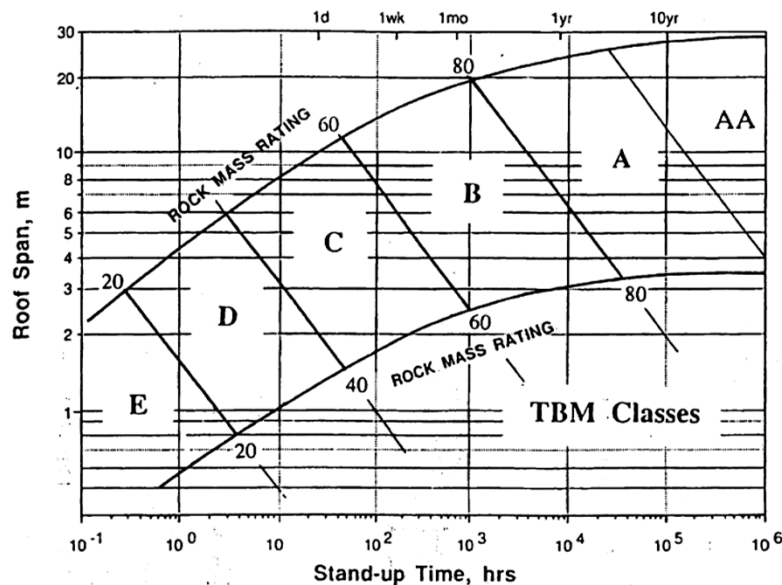


Figure 4.2 Modified 1988 Lauffer diagram depicting boundaries of rock mass classes for TBM applications. (After Lauffer, 1988.)

Design Methods – Empirical Approach

Geomechanics Classification

Rock Load height

The concept of rock load height, H_t , was first proposed by Terzaghi (1946). The rock load height (the height of rock which should be supported) can be related to the rock mass rating using the below Equation (Unal, 1983)

$$H_t = (100 - \text{RMR}) \times (2W) / 100$$

Where; H_t is the rock load height, m;

$2W$ is the width of the excavation span, m; and

RMR is the rock mass rating.

Design Methods – Empirical Approach

Geomechanics Classification

Rock Load

Once rock load height has been determined the rock load, P_r , can be found using Equation

$$P_r = \gamma_r H_t$$

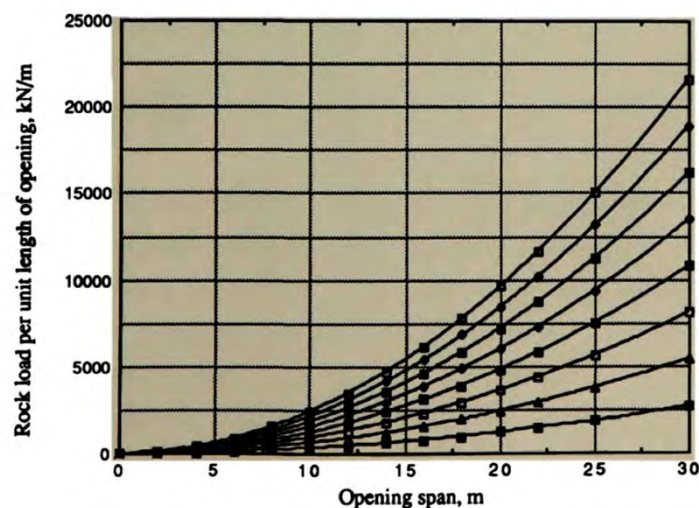
Where; P_r is the rock load per unit length of opening, kN/m; and

γ_r is the unit weight of rock material, kN/m³.

$$P_r = (100 - \text{RMR}) \times (2W) \times \gamma_r / 100$$

Design Methods – Empirical Approach

Geomechanics Classification



Rock load vs roof span in various rock classes (after Unal, 1983)

Design Methods – Empirical Approach

THE Q-SYSTEM

The Q-value gives a description of the rock mass stability of an underground opening in jointed rock masses.

High Q-values indicates good stability and low values means poor stability.

Based on 6 parameters the Q-value is calculated using the following equation:

$$Q = \frac{RQD}{J_n} \times \frac{J_r}{J_a} \times \frac{J_w}{SRF}$$

Design Methods – Empirical Approach

THE Q-SYSTEM

$$Q = \frac{RQD}{J_n} \times \frac{J_r}{J_a} \times \frac{J_w}{SRF}$$

The six parameters are:

RQD = Degree of jointing (Rock Quality Designation)

J_n = Number of joint sets

J_r = Joint roughness number

J_a = Joint alteration number

J_w = Joint water reduction factor

SRF = Stress Reduction Factor

Design Methods – Empirical Approach

THE Q-SYSTEM

$$Q = \frac{RQD}{J_n} \times \frac{J_r}{J_a} \times \frac{J_w}{SRF}$$

$$\frac{RQD}{J_n} = \text{Degree of jointing (or block size)}$$

$$\frac{J_r}{J_a} = \text{Joint friction (inter-block shear strength)}$$

$$\frac{J_w}{SRF} = \text{Active stress}$$

7 Type of excavation		ESR
A	Temporary mine openings, etc.	ca. 3-5
B	Vertical shafts*: i) circular sections ii) rectangular/square section * Dependant of purpose. May be lower than given values.	ca. 2.5 ca. 2.0
C	Permanent mine openings, water tunnels for hydro power (exclude high pressure penstocks) water supply tunnels, pilot tunnels, drifts and headings for large openings.	1.6
D	Minor road and railway tunnels, surge chambers, access tunnels, sewage tunnels, etc.	1.3
E	Power houses, storage rooms, water treatment plants, major road and railway tunnels, civil defence chambers, portals, intersections, etc.	1.0
F	Underground nuclear power stations, railways stations, sports and public facilities, factories, etc.	0.8
G	Very important caverns and underground openings with a long lifetime, ≈ 100 years, or without access for maintenance.	0.5

Design Methods – Empirical Approach

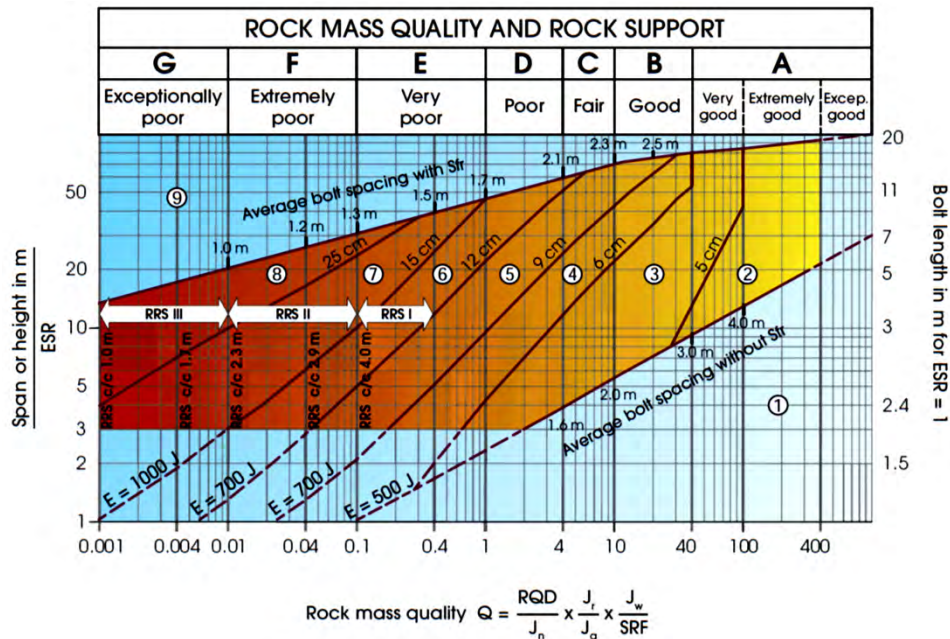
USING THE Q-SYSTEM TO EVALUATE THE SUPPORT REQUIREMENTS

Excavation Support Ratio (ESR)

It is recommended to use $ESR = 1.0$ when $Q \leq 0.1$ for the types of excavation B, C and D. The reason for that is that the stability problems may be severe with such low Q-values, perhaps with risk for cave-in.

In addition to the span (or wall height) ESR gives the “Equivalent dimension” in the following way:

$$\frac{\text{Span or height in m}}{ESR} = \text{Equivalent dimension}$$



Support categories

- ① Unsupported or spot bolting
- ② Spot bolting, **SB**
- ③ Systematic bolting, fibre reinforced sprayed concrete, 5-6 cm, **B+Sfr**
- ④ Fibre reinforced sprayed concrete and bolting, 6-9 cm, **Sfr (E500)+B**
- ⑤ Fibre reinforced sprayed concrete and bolting, 9-12 cm, **Sfr (E700)+B**
- ⑥ Fibre reinforced sprayed concrete and bolting, 12-15 cm + reinforced ribs of sprayed concrete and bolting, **Sfr (E700)+RRS I+B**
- ⑦ Fibre reinforced sprayed concrete >15 cm + reinforced ribs of sprayed concrete and bolting, **Sfr (E1000)+RRS II+B**
- ⑧ Cast concrete lining, **CCA** or **Sfr (E1000)+RRS III+B**
- ⑨ Special evaluation

Bolts spacing is mainly based on Ø20 mm

E = Energy absorption in fibre reinforced sprayed concrete

ESR = Excavation Support Ratio

Areas with dashed lines have no empirical data

RRS - spacing related to Q-value

I **Si30/6 Ø16 - Ø20 (span 10m)**
D40/6+2 Ø16-20 (span 20m)

II **Si35/6 Ø16-20 (span 5m)**
D45/6+2 Ø16-20 (span 10m)
D55/6+4 Ø20 (span 20m)

III **D40/6+4 Ø16-20 (span 5 m)**
D55/6+4 Ø20 (span 10 m)
D70/6+6 Ø20 (span 20 m)

Si30/6 = Single layer of 6 rebars,
30 cm thickness of sprayed concrete

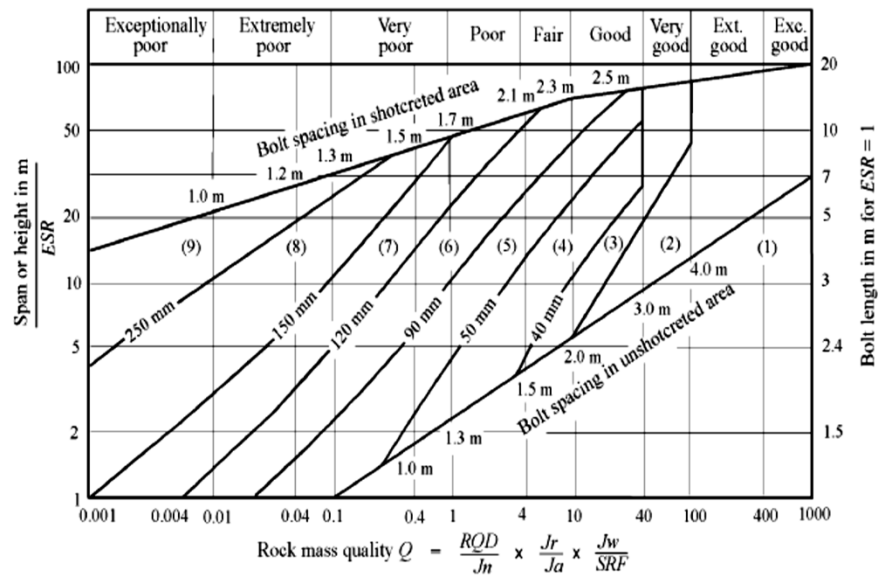
D = Double layer of rebars

Ø16 = Rebar diameter is 16 mm

c/c = RSS spacing, centre - centre

In the legend belonging to the support chart, the following notations are used:

- "Si" means single layer of steel bars
- "D" means double layers of steel bars
- "45" means a total rib thickness of 45 cm
- "6" means 6 steel bars
- "c/c = 2-3" means a centre to centre spacing of 2 to 3 metres between the ribs
- "16" or "20" gives the diameter of the steel bar in mm



The length of the bolts depends on the span or wall height of the underground opening and to some degree on the rock mass quality.

In unfavourable joint geometry, longer bolts than recommended in the diagram will be necessary, and there is also a general need for increasing bolt length by decreasing Q-value.

Wall support

The support chart is primarily valid for the crown and springlines of underground openings and caverns. The level of support on the walls is normally less for high and intermediate Q-values ($Q > 0.1$).

When the Q-system is used for wall support, the height of the walls must be used instead of the span.

Table 8 Conversion from actual Q-values to adjusted Q-values for design of wall support.

In rock masses of good quality	$Q > 10$	Multiply Q-values by a factor of 5.
For rock masses of intermediate quality	$0.1 < Q < 10$	Multiply Q-values by a factor of 2.5. In cases of high rock stresses, use the actual Q-value.
For rock masses of poor quality	$Q < 0.1$	Use actual Q-value.

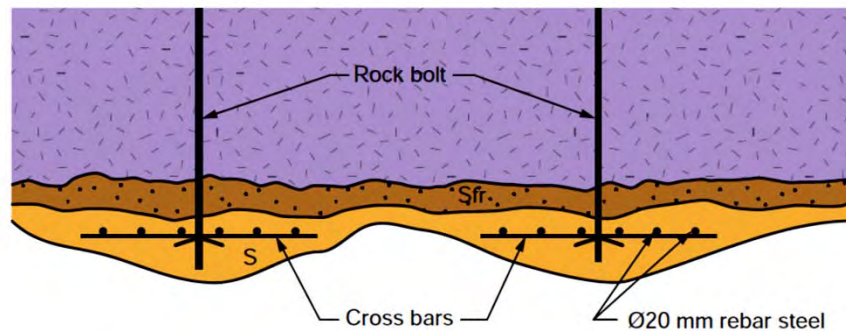
USING THE Q-SYSTEM TO EVALUATE THE SUPPORT REQUIREMENTS

Reinforced ribs of sprayed concrete (RRS)

In sections with very poor rock mass quality ($Q < 1$), reinforced ribs of sprayed concrete (RRS) in many cases is a preferred alternative to cast concrete.

The ribs are constructed with a combination of steel bars (usually with a diameter of 16 mm or 20 mm), sprayed concrete and rock bolts

Principle of construction of RRS.



In cases where the Q-values indicates use of RRS, a 12-15 cm thick layer of fibre reinforced sprayed concrete normally has to be applied before installing the reinforced ribs. This layer has a function as temporary support as well as to smooth out the rock surface. The thickness of this layer is included in the total thickness of the RRS.

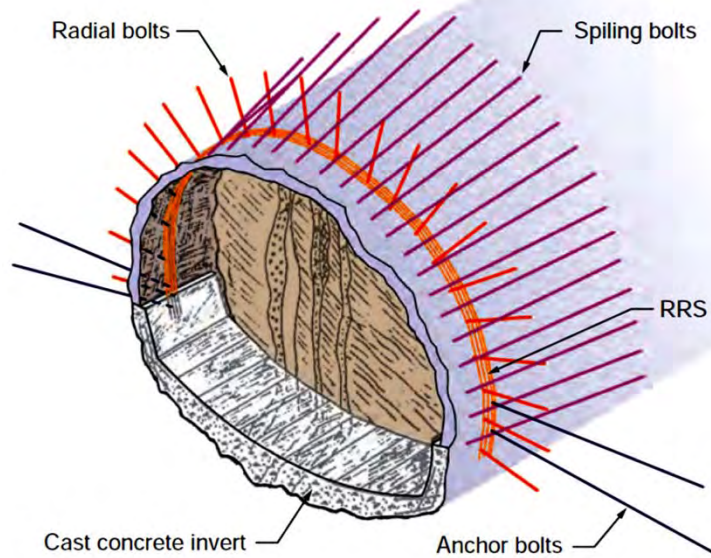
USING THE Q-SYSTEM TO EVALUATE THE SUPPORT REQUIREMENTS

FOREPOLING

- In poor rock mass qualities it may be necessary to forepole, i.e. by installation of spiling bolts in front of the working face in order to avoid overbreak or cave in. Forepoling is not directly included in the support chart related to the Q-system.
- However, the suggested rock support in rock masses with low Q-values is based on the use of forepoling during excavation.

FOREPOLING

- The need for forepoling is dependent on the span of the underground opening and the rock mass quality.
- Generally, it is suggested to use forepoling in rock masses with Q-values lower than 0.1–0.6, depending on the span of the underground opening.
- The spacing between the forepoles is normally around 0.3 m (0.2 - 0.6 m). The rear end of the forepoling has to be anchored in the overlying rock connected to radial rock bolts combined with steel straps or rebar steel and sprayed concrete in order to avoid collapse during excavation.



Support of poor rock masses by spiling bolts

RMR Support Design

$$L = \frac{2 + 0.15B}{ESR}$$

$$P_{RMR} = \frac{100 - RMR}{100} B \cdot \rho$$

$$n = \frac{B \cdot ht \cdot c \cdot \rho \cdot SF}{R_{max}}$$

$$t = 0.434 \frac{P_{RMR} \cdot B}{\tau} \text{ and } \tau = 0.2 \frac{LB}{SF}$$

Where:

L = rock bolt length (m)

B = excavation width (m)

ESR = Excavation Support Ratio

n = number of rock bolt (pieces)

ht = rock-load height (m)

c = span excavation (m)

ρ = rock density (ton/m³)

SF = safety factor

R_{max} = bearing capacity (ton)

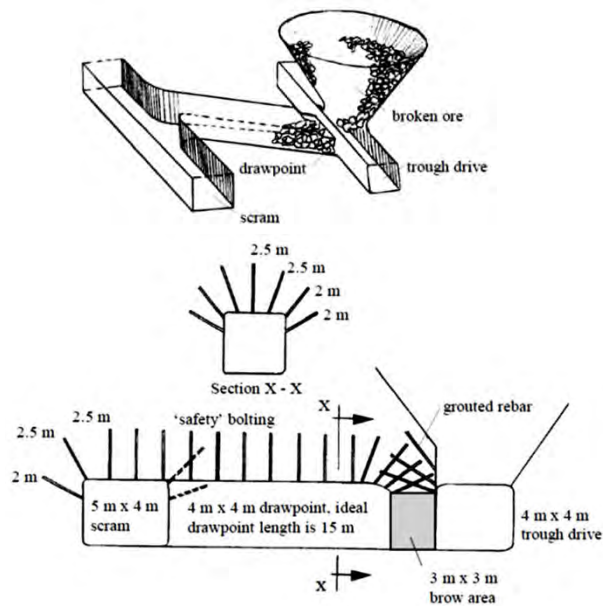
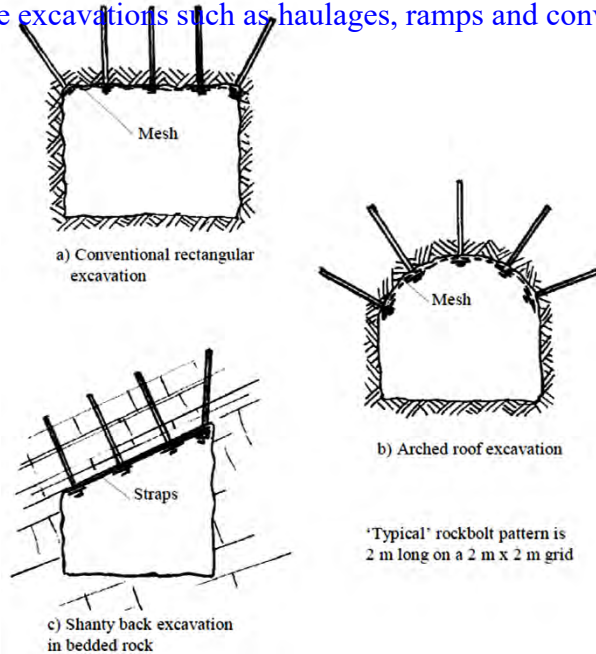
LB = UCS shotcrete (ton/m²)

t = shotcrete thickness

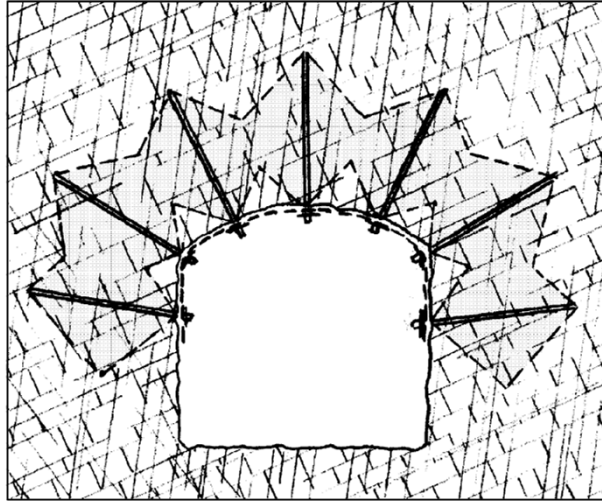
τ = shotcrete shear stress (ton/m²)

P_{RMR} = rock-load pressure (ton/m²)

Typical support systems which can be installed to improve the safety of service excavations such as haulages, ramps and conveyor tunnels



Suggested reinforcement for a drawpoint in a large mechanised mine. The brow area, shown shaded, is blasted last after the rebar has been grouted in place from the drawpoint and trough drive. 'Safety' bolting can be used in the drawpoint and scram.

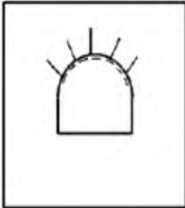
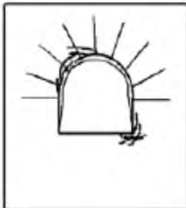


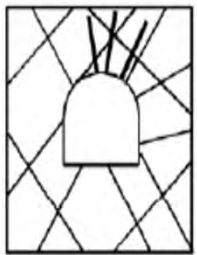
Pattern bolting for the support of heavily jointed rock which can fail by ravelling or squeezing. The shading represents a zone of compressive stress in which interlocking of individual rock pieces is retained and a self-supporting arch is created. Mesh or shotcrete should be applied to the excavation surface to retain small blocks and wedges in the stress-free zones between the rockbolts.

	Low stress levels	High stress levels
Massive rock	 Massive rock subjected to low in situ stress levels. No support or 'safety bolts' or dowels and mesh.	 Massive rock subjected to high in situ stress levels. Pattern rockbolts or dowels with mesh or shotcrete to inhibit fracturing and to keep broken rock in place.
Jointed rock	 Massive rock with relatively few discontinuities subjected to low in situ stress conditions. 'Spot' bolts located to prevent failure of individual blocks and wedges. Bolts must be tensioned.	 Massive rock with relatively few discontinuities subjected to high in situ stress conditions. Heavy bolts or dowels, inclined to cross rock structure, with mesh or steel fibre reinforced shotcrete on roof and side walls.
Heavily jointed rock	 Heavily jointed rock subjected to low in situ stress conditions. Light pattern bolts with mesh and/or shotcrete will control ravelling of new surface rock pieces.	 Heavily jointed rock subjected to high in situ stress conditions. Heavy rockbolt or dowel pattern with steel fibre reinforced shotcrete. In extreme cases, steel sets with sliding joints may be required. Invert struts or concrete floor slabs may be required to control floor heave.

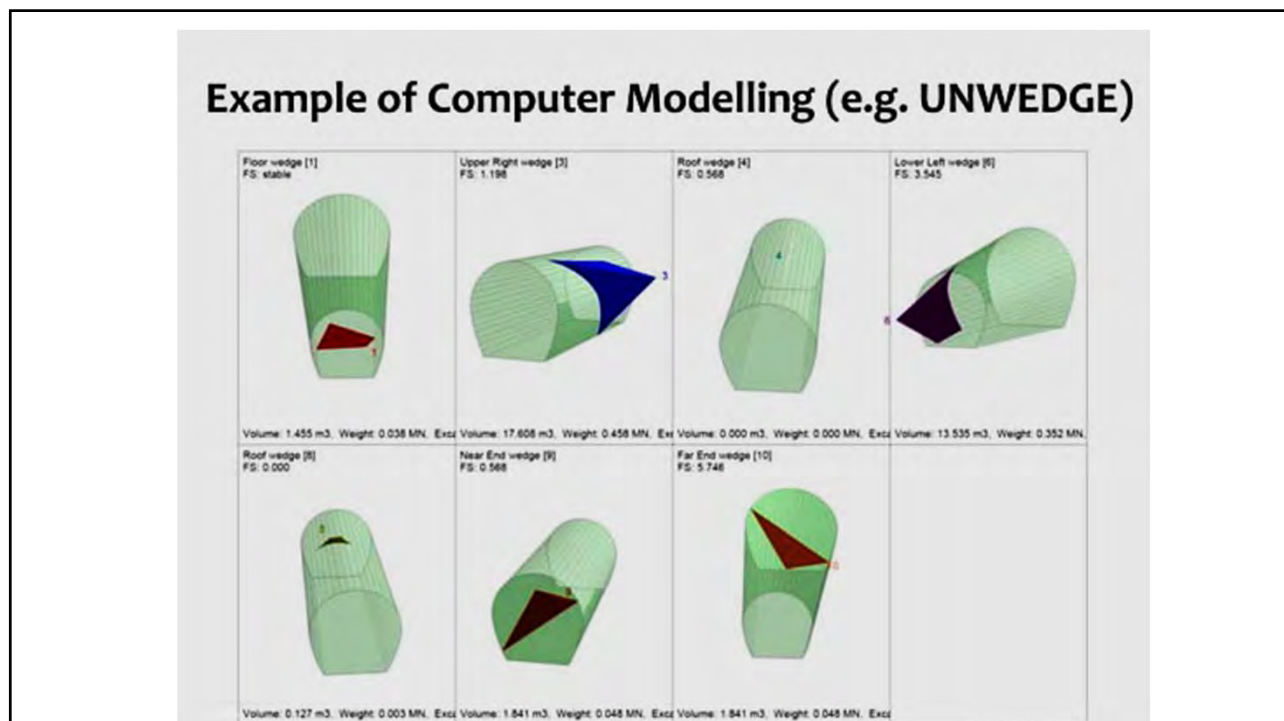
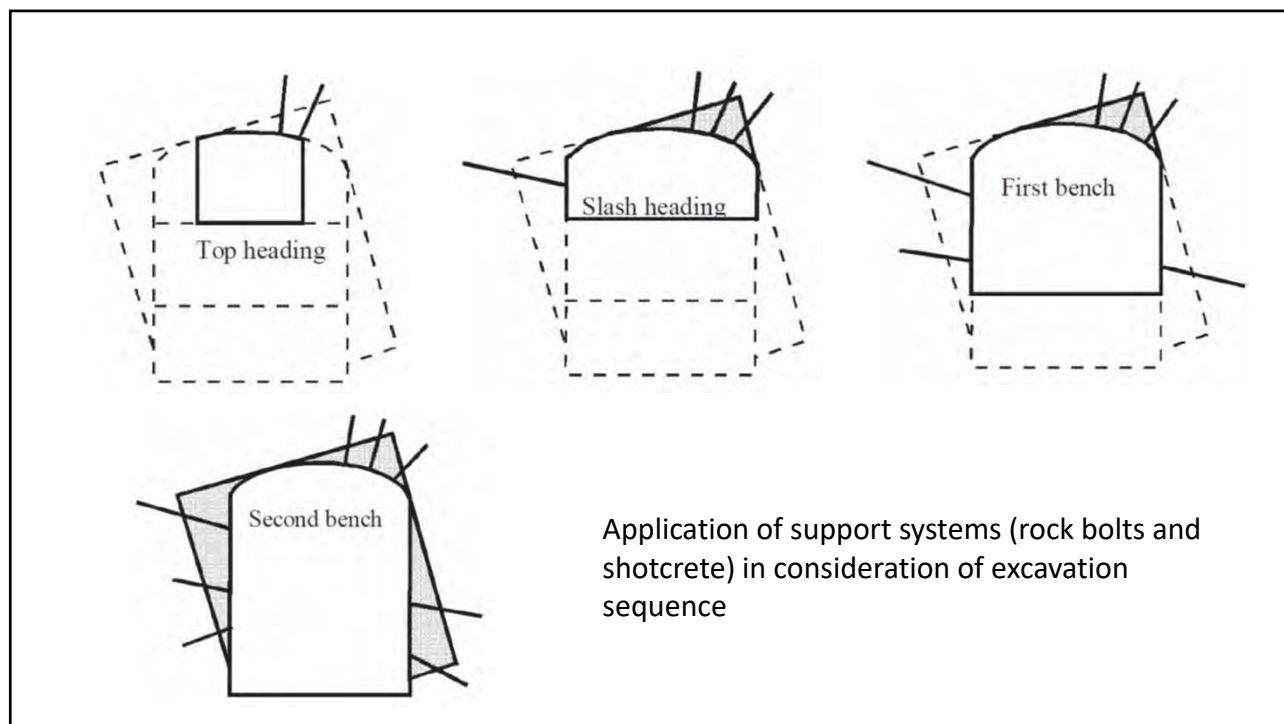
Typical rockbolt and dowel applications to control different types of rock mass failure

		Low stress levels	High stress levels	Typical rockbolt and dowel applications to control different types of rock mass failure
Massive rock		Massive rock subjected to low in situ stress levels. Linear elastic response with little or no rock failure.		
Jointed rock		Massive rock, with relatively few discontinuities, subjected to low in situ stress conditions. Blocks or wedges, released by intersecting discontinuities, fall or slide due to gravity loading.		
Heavily jointed rock		Heavily jointed rock subjected to low in situ stress conditions. The opening surface falls as a result of unravelling of small interlocking blocks and wedges. Failure can propagate a long way into the rock mass if it is not controlled.		

	Low stress levels	High stress levels
Massive rock	 Massive rock subjected to low in situ stress levels. No permanent support. Light support may be required for construction safety.	 Massive rock subjected to high in situ stress levels. Pattern rockbolts or dowels with mesh or shotcrete to inhibit fracturing and to keep broken rock in place.

		Low stress levels	High stress levels	
	Jointed rock	 Massive rock with relatively few discontinuities subjected to low in situ stress conditions. 'Spot' bolts located to prevent failure of individual blocks and wedges. Bolts must be tensioned.	 Massive rock with relatively few discontinuities subjected to high in situ stress conditions. Heavy bolts or dowels, inclined to cross rock structure, with mesh or steel fibre reinforced shotcrete on roof and sidewalls.	

		Low stress levels	High stress levels	
	Heavily jointed rock	 Heavily jointed rock subjected to low in situ stress conditions. Light pattern bolts with mesh and/or shotcrete will control ravelling of near surface rock pieces.	 Heavily jointed rock subjected to high in situ stress conditions. Heavy rockbolt or dowel pattern with steel fibre reinforced shotcrete. In extreme cases, steel sets with sliding joints may be required. Invert struts or concrete floor slabs may be required to control floor heave.	

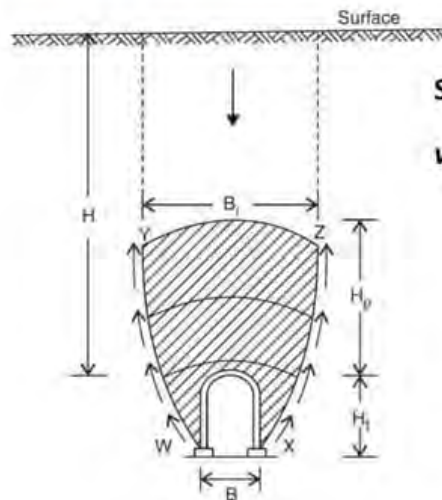


Estimate Rock Support Force for Rock Tunnels and Caverns

Estimation of Design Rock Load

- **Terzaghi Rock Load**
(Terzaghi 1946, Deere et al. 1970; Singh et al. 1995)
- **RMR Support Pressure**
(Bieniawski 1984, Unal 1983, Goel and Jethwa 1991)
- **Q Support Pressure**
(Barton et al. 1974, 1975, Grimstad and Barton 1993)
- **Elastic Solutions (e.g. Kirsch Equations)** for circular opening
- **Numerical Modelling** (e.g. FLAC, UDEC, PHASE²)

Terzaghi Rock Load



Support pressure (p_v) = $\gamma \cdot H_p$
 where γ is unit weight of rock

Terzaghi Rock Load

Terzaghi's classification			Classification of Singh et al. (1995)				
Category	Rock condition	Rock load factor H_p	Category	Rock condition	Recommended support pressure (MPa)		Remarks
					p_v	p_h	
I.	Hard and intact	0	I.	Hard and intact	0	0	—
II.	Hard stratified or schistose	0 to 0.5B	II.	Hard stratified or schistose	0.04-0.07	0	—
III.	Massive, moderately jointed	0 to 0.25B	III.	Massive, moderately jointed	0.0-0.04	0	—
IV.	Moderately blocky, seamy and jointed	0.25B to 0.35 (B + H _t)	IV.	Moderately blocky seamy very jointed	0.04-0.1	0-0.2 p_v	Inverts may be required
V.	Very blocky and seamy, shattered, arched	0.35 to 1.1 (B + H _t)	V.	Very blocky and seamy, shattered highly jointed, thin shear zone or fault	0.1-0.2	0-0.5 p_v	Inverts may be required, arched roof preferred
VI.	Completely crushed but chemically intact	1.1 (B + H _t)	VI.	Completely crushed but chemically unaltered, thick shear and fault zone	0.2-0.3	0.3-1.0 p_v	Inverts essential, arched roof essential

(Singh and Goel 2006)

Terzaghi Rock Load

Comments on Terzaghi Rock Load

- Terzaghi's method provides reasonable support pressure for small tunnels ($B < 6$ m).
- It provides over-safe estimates for large tunnels and caverns (Diam. 6 to 14 m) and
- The estimated support pressure values fall in a very large range for squeezing and swelling ground conditions for a meaningful application.

Comments on Terzaghi Rock Load (cont')

Barton et al. (1974) and Verman (1993) suggested that the support pressure is independent of opening width in rock tunnels. Goel et al. (1996) also found that there is a negligible effect of tunnel size on support pressure in non-squeezing ground conditions, but the tunnel size could have considerable influence on the support pressure in squeezing ground condition.

RMR Support Pressure

$$p_v = \left[\frac{100 - \text{RMR}}{100} \right] \cdot \gamma \cdot B$$

Unal (1983) for flat roof coal mine

$$p_v = \frac{7.5 B^{0.1} \cdot H^{0.5} - \text{RMR}}{20 \text{ RMR}}$$

Goel and Jethwa (1991) short-term support pressure for underground openings, but not for rock burst condition

where H = overburden or tunnel depth in meters (50–600 m)

Q Support Pressure

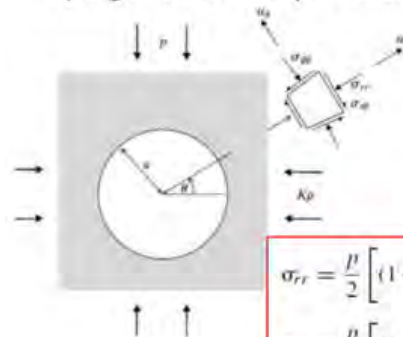
Barton et al. 1974 recommended:

$$P_v = \frac{200 \cdot Q^{-1/3}}{J_r} \quad (\text{kPa}), \text{ if no. of joint sets} \geq 3$$

$$P_v = \frac{200 \cdot J_n^{1/2} \cdot Q^{-1/3}}{3 \cdot J_r} \quad (\text{kPa}), \text{ if no. of joint sets} \leq 3$$

Elastic Solutions

(e.g. Kirsch Equations for circular opening)



$$\sigma_{rr} = \frac{p}{2} \left[(1+K) \left(1 - \frac{a^2}{r^2} \right) - (1-K) \left(1 - 4\frac{a^2}{r^2} + \frac{3a^4}{r^4} \right) \cos 2\theta \right]$$

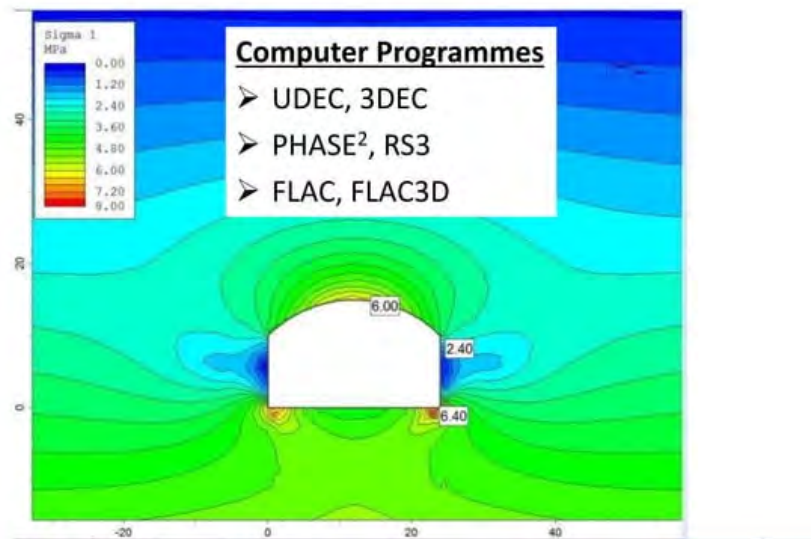
$$\sigma_{\theta\theta} = \frac{p}{2} \left[(1+K) \left(1 + \frac{a^2}{r^2} \right) + (1-K) \left(1 + \frac{3a^4}{r^4} \right) \cos 2\theta \right]$$

$$\sigma_{r\theta} = \frac{p}{2} \left[(1-K) \left(1 + \frac{2a^2}{r^2} - \frac{3a^4}{r^4} \right) \sin 2\theta \right]$$

$$u_r = -\frac{pa^2}{4Gr} \left[(1+K) - (1-K) \left\{ 4(1-\nu) - \frac{a^2}{r^2} \right\} \cos 2\theta \right]$$

$$u_\theta = -\frac{pa^2}{4Gr} \left[(1-K) \left\{ 2(1-2\nu) + \frac{a^2}{r^2} \right\} \sin 2\theta \right]$$

Numerical Modelling



Q-system Approach Rock Support Design

Tunnel/Cavern Support Design in Rock

Q-system Design Approach:

- 1) Determine an opening size (height, span)
- 2) Determine Excavation Support Ratio (ESR)

Excavation category		ESR
A	Temporary mine openings.	3-5
B	Permanent mine openings, water tunnels for hydro power (excluding high pressure penstocks), pilot tunnels, drifts and headings for large excavations.	1.6
C	Storage rooms, water treatment plants, minor road and railway tunnels, surge chambers, access tunnels.	1.3
D	Power stations, major road and railway tunnels, civil defence chambers, portal intersections.	1.0
E	Underground nuclear power stations, railway stations, sports and public facilities, factories.	0.8

Q-system Design Approach (cont'):

- 3) Calculate roof and wall supporting stress for different Q-values

$$P_v = \frac{200 \cdot Q^{-1/3}}{J_r} \quad (\text{kPa}), \text{ if no. of joint sets} \geq 3$$

$$P_v = \frac{200 \cdot J_n^{1/2} \cdot Q^{-1/3}}{3 \cdot J_r} \quad (\text{kPa}), \text{ if no. of joint sets} \leq 3$$

Range of Q	Wall		Roof
	Factored Q_{wall}	Temporary Q^t_{wall}	Temporary Q^t_{roof}
$Q > 10$	$5.0 \times Q$	$5 \times 5 \times Q$	$5.0 \times Q$
$0.1 < Q < 10$	$2.5 \times Q$	$5 \times 2.5 \times Q$	$5.0 \times Q$
$Q < 0.1$	$1.0 \times Q$	$5 \times 1.0 \times Q$	$5.0 \times Q$

Q-system Design Approach (cont'):

- 4) Determine bolt length, bolt force and bolt spacing

$$L = 2 + \frac{0.15B}{ESR}$$

Bolt length (m). B is span or height of opening whichever is larger.
For temporary case ($< 1\text{yr}$), $ESR_{\text{temp}} = ESR \times 1.5$

Design Working Load of Bolt $\leq 0.5 \times$ characteristic yield strength of bolt (e.g. BS 8081)

$$\text{Bolt Spacing} = (\text{Support pressure} / \text{Working Load of Bolt})^{0.5}$$

Q-system Design Approach (cont'):

